WEEK 3 – PROCESS SCHEDULING SIMULATOR

Project Report

1. Objective

To design and implement a CPU scheduling simulator in C that simulates different CPU scheduling algorithms and analyzes their performance.

The simulator accepts process details, displays Gantt charts, calculates scheduling metrics, and evaluates CPU utilization.

2. Input Specification

The CPU Scheduling Simulator takes process details as input and executes the selected CPU scheduling algorithm. These inputs are used to calculate Waiting Time (WT), Turnaround Time (TAT), Response Time (RT), and their averages.

1. Number of Processes (n):

   Specifies the total number of processes to be scheduled.

2. Process ID (PID):

   A unique identifier assigned to each process, such as P1, P2, P3, and P4.

3. Arrival Time (AT):

   The time at which a process enters the ready queue.

4. Burst Time (BT):

   The total CPU execution time required by a process.

5. Priority:

   Indicates the priority of a process. A smaller number represents a higher priority.

6. Time Quantum:

   The fixed time slice allocated to each process in the Round Robin algorithm.

7. Scheduling Algorithm:

   The user selects one of the six implemented algorithms:

- FCFS (First Come First Serve)
  - SJF (Shortest Job First - Non-preemptive)
  - Priority Scheduling (Non-preemptive)
  - SRTF (Shortest Remaining Time First)
  - Priority Scheduling (Preemptive)
  - Round Robin (RR)

3. Performance Metrics

Completion Time (CT): The time when a process finishes execution.

Turnaround Time (TAT):
TAT = CT − AT

Waiting Time (WT):
WT = TAT − BT

Response Time (RT):
RT = First CPU Start Time − AT

Average Waiting Time:
Average WT = Total Waiting Time ÷ Number of Processes

Average Turnaround Time:
Average TAT = Total Turnaround Time ÷ Number of Processes

Average Response Time:
Average RT = Total Response Time ÷ Number of Processes

CPU Utilization:
CPU Utilization (%) = (CPU Busy Time ÷ Total Elapsed Time) × 100
CPU Busy Time = Total Elapsed Time − Total Idle Time

If the CPU is busy continuously from time 0 until all processes finish, CPU utilization is 100%. If idle intervals occur, include them in the calculation.

4. Test Case 1

```
========== CPU SCHEDULING SIMULATOR ==========
Enter number of processes (1-100): 3

Process 1
PID: 1
Arrival time: 0
Burst time: 7
Priority: 3

Process 2
PID: 2
Arrival time: 1
Burst time: 5
Priority: 1

Process 3
PID: 3
Arrival time: 2
Burst time: 2
Priority: 4

Enter Round Robin time quantum (> 0): 2
```

4.1 FCFS

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 1

========== GANTT CHART ==========

| P1 | P2 | P3 |
0   7    12   14

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     7     3         7      7       0      0
2      1     5     1         12     11      6      6
3      2     2     4         14     12      10     10

========== AVERAGES ==========
Average Waiting Time    : 5.33
Average Turnaround Time : 10.00
Average Response Time   : 5.33

========== CPU UTILIZATION ==========
CPU Busy Time  : 14
CPU Idle Time  : 0
Total Time     : 14
CPU Utilization: 100.00%
```

4.2 SJF Non-preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 2

========== GANTT CHART ==========

| P1 | P3 | P2 |
0   7    9    14

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     7     3         7      7       0      0
2      1     5     1         14     13      8      8
3      2     2     4         9      7       5      5

========== AVERAGES ==========
Average Waiting Time    : 4.33
Average Turnaround Time : 9.00
Average Response Time   : 4.33

========== CPU UTILIZATION ==========
CPU Busy Time  : 14
CPU Idle Time  : 0
Total Time     : 14
CPU Utilization: 100.00%
```

4.3 SJF Preemptive (SRTF)

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 3

========== GANTT CHART ==========

| P1 | P2 | P3 |
0   7    12   14

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     7     3         7      7       0      0
2      1     5     1         12     11      6      6
3      2     2     4         14     12      10     10

========== AVERAGES ==========
Average Waiting Time    : 5.33
Average Turnaround Time : 10.00
Average Response Time   : 5.33

========== CPU UTILIZATION ==========
CPU Busy Time  : 14
CPU Idle Time  : 0
Total Time     : 14
CPU Utilization: 100.00%
```

4.4 Priority Non-preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 4

========== GANTT CHART ==========

| P1 | P2 | P3 | P2 | P1 |
0   1    2    4    8    14

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     7     3         14     14      7      0
2      1     5     1         8      7       2      0
3      2     2     4         4      2       0      0

========== AVERAGES ==========
Average Waiting Time    : 3.00
Average Turnaround Time : 7.67
Average Response Time   : 0.00

========== CPU UTILIZATION ==========
CPU Busy Time  : 14
CPU Idle Time  : 0
Total Time     : 14
CPU Utilization: 100.00%
```

4.5 Priority Preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 5

========== GANTT CHART ==========

| P1 | P2 | P1 | P3 |
0   1    6    12   14

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     7     3         12     12      5      0
2      1     5     1         6      5       0      0
3      2     2     4         14     12      10     10

========== AVERAGES ==========
Average Waiting Time    : 5.00
Average Turnaround Time : 9.67
Average Response Time   : 3.33

========== CPU UTILIZATION ==========
CPU Busy Time  : 14
CPU Idle Time  : 0
Total Time     : 14
CPU Utilization: 100.00%
```

4.6 Round Robin

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 6

========== GANTT CHART ==========

| P1 | P2 | P3 | P1 | P2 | P1 | P2 | P1 |
0   2    4    6    8    10   12   13   14

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     7     3         14     14      7      0
2      1     5     1         13     12      7      1
3      2     2     4         6      4       2      2

========== AVERAGES ==========
Average Waiting Time    : 5.33
Average Turnaround Time : 10.00
Average Response Time   : 1.00

========== CPU UTILIZATION ==========
CPU Busy Time  : 14
CPU Idle Time  : 0
Total Time     : 14
CPU Utilization: 100.00%
```

4.7 Compare All Algorithm

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 7

================ ALGORITHM COMPARISON ================
Algorithm                      Avg WT      Avg TAT       Avg RT    CPU Usage
FCFS                             5.33        10.00         5.33      100.00%
SJF Non-Preemptive               4.33         9.00         4.33      100.00%
Priority Non-Preemptive          5.33        10.00         5.33      100.00%
SRTF                             3.00         7.67         0.00      100.00%
Priority Preemptive              5.00         9.67         3.33      100.00%
Round Robin                      5.33        10.00         1.00      100.00%

Round Robin Time Quantum: 2
```

5. Test Case 2

```
========== CPU SCHEDULING SIMULATOR ==========
Enter number of processes (1-100): 4

Process 1
PID: 1
Arrival time: 0
Burst time: 4
Priority: 2

Process 2
PID: 2
Arrival time: 0
Burst time: 7
Priority: 1

Process 3
PID: 3
Arrival time: 3
Burst time: 2
Priority: 3

Process 4
PID: 4
Arrival time: 5
Burst time: 3
Priority: 2

Enter Round Robin time quantum (> 0): 2
```

5.1 FCFS

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 1

========== GANTT CHART ==========

| P1 | P2 | P3 | P4 |
0   4    11   13   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     4     2         4      4       0      0
2      0     7     1         11     11      4      4
3      3     2     3         13     10      8      8
4      5     3     2         16     11      8      8

========== AVERAGES ==========
Average Waiting Time    : 5.00
Average Turnaround Time : 9.00
Average Response Time   : 5.00

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

5.2 SJF Non-preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 2

========== GANTT CHART ==========

| P1 | P3 | P4 | P2 |
0   4    6    9    16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     4     2         4      4       0      0
2      0     7     1         16     16      9      9
3      3     2     3         6      3       1      1
4      5     3     2         9      4       1      1

========== AVERAGES ==========
Average Waiting Time    : 2.75
Average Turnaround Time : 6.75
Average Response Time   : 2.75

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

5.3 SJF Preemptive(SRTF)

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 3

========== GANTT CHART ==========

| P2 | P1 | P4 | P3 |
0   7    11   14   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     4     2         11     11      7      7
2      0     7     1         7      7       0      0
3      3     2     3         16     13      11     11
4      5     3     2         14     9       6      6

========== AVERAGES ==========
Average Waiting Time    : 6.00
Average Turnaround Time : 10.00
Average Response Time   : 6.00

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

5.4 Priority Non-Preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 4

========== GANTT CHART ==========

| P1 | P3 | P4 | P2 |
0   4    6    9    16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     4     2         4      4       0      0
2      0     7     1         16     16      9      9
3      3     2     3         6      3       1      1
4      5     3     2         9      4       1      1

========== AVERAGES ==========
Average Waiting Time    : 2.75
Average Turnaround Time : 6.75
Average Response Time   : 2.75

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

5.5 Priority Preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 5

========== GANTT CHART ==========

| P2 | P1 | P4 | P3 |
0   7    11   14   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     4     2         11     11      7      7
2      0     7     1         7      7       0      0
3      3     2     3         16     13      11     11
4      5     3     2         14     9       6      6

========== AVERAGES ==========
Average Waiting Time    : 6.00
Average Turnaround Time : 10.00
Average Response Time   : 6.00

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

5.6 Round Robin

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 6

========== GANTT CHART ==========

| P1 | P2 | P1 | P3 | P2 | P4 | P2 | P4 | P2 |
0   2    4    6    8    10   12   14   15   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      0     4     2         6      6       2      0
2      0     7     1         16     16      9      2
3      3     2     3         8      5       3      3
4      5     3     2         15     10      7      5

========== AVERAGES ==========
Average Waiting Time    : 5.25
Average Turnaround Time : 9.25
Average Response Time   : 2.50

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

5.7 Compare All Algorithm

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 7

================ ALGORITHM COMPARISON ================
Algorithm                      Avg WT      Avg TAT      Avg RT    CPU Usage
FCFS                             5.00         9.00        5.00      100.00%
SJF Non-Preemptive               2.75         6.75        2.75      100.00%
Priority Non-Preemptive          6.00        10.00        6.00      100.00%
SRTF                             2.75         6.75        2.75      100.00%
Priority Preemptive              6.00        10.00        6.00      100.00%
Round Robin                      5.25         9.25        2.50      100.00%

Round Robin Time Quantum: 2
```

6. Test Case 3

```
========== CPU SCHEDULING SIMULATOR ==========
Enter number of processes (1-100): 4

Process 1
PID: 1
Arrival time: 2
Burst time: 3
Priority: 2

Process 2
PID: 2
Arrival time: 4
Burst time: 6
Priority: 1

Process 3
PID: 3
Arrival time: 7
Burst time: 2
Priority: 3

Process 4
PID: 4
Arrival time: 0
Burst time: 5
Priority: 2

Enter Round Robin time quantum (> 0): 2
```

6.1 FCFS

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 1

========== GANTT CHART ==========

| P4 | P1 | P2 | P3 |
0   5    8    14   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      2     3     2         8      6       3      3
2      4     6     1         14     10      4      4
3      7     2     3         16     9       7      7
4      0     5     2         5      5       0      0

========== AVERAGES ==========
Average Waiting Time    : 3.50
Average Turnaround Time : 7.50
Average Response Time   : 3.50

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

6.2 SJF Non-preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 2

========== GANTT CHART ==========

| P4 | P1 | P3 | P2 |
0   5    8    10   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      2     3     2         8      6       3      3
2      4     6     1         16     12      6      6
3      7     2     3         10     3       1      1
4      0     5     2         5      5       0      0

========== AVERAGES ==========
Average Waiting Time    : 2.50
Average Turnaround Time : 6.50
Average Response Time   : 2.50

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

6.3 SJF Preemptive(SRTF)

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 2

========== GANTT CHART ==========

| P4 | P1 | P3 | P2 |
0   5    8    10   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      2     3     2         8      6       3      3
2      4     6     1         16     12      6      6
3      7     2     3         10     3       1      1
4      0     5     2         5      5       0      0

========== AVERAGES ==========
Average Waiting Time    : 2.50
Average Turnaround Time : 6.50
Average Response Time   : 2.50

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

6.4 Priority Non-preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 4

========== GANTT CHART ==========

| P4 | P1 | P3 | P2 |
0   5    8    10   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      2     3     2         8      6       3      3
2      4     6     1         16     12      6      6
3      7     2     3         10     3       1      1
4      0     5     2         5      5       0      0

========== AVERAGES ==========
Average Waiting Time    : 2.50
Average Turnaround Time : 6.50
Average Response Time   : 2.50

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

6.5 Priority Preemptive

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 5

========== GANTT CHART ==========

| P4 | P2 | P1 | P4 | P3 |
0   4    10   13   14   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      2     3     2         13     11      8      8
2      4     6     1         10     6       0      0
3      7     2     3         16     9       7      7
4      0     5     2         14     14      9      0

========== AVERAGES ==========
Average Waiting Time    : 6.00
Average Turnaround Time : 10.00
Average Response Time   : 3.75

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

6.6 Round Robin

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 6

========== GANTT CHART ==========

| P4 | P1 | P4 | P2 | P1 | P4 | P3 | P2 |
0   2    4    6    8    9    10   12   16

================ PROCESS RESULTS ================
PID    AT    BT    Priority  CT     TAT     WT     RT
1      2     3     2         9      7       4      0
2      4     6     1         16     12      6      2
3      7     2     3         12     5       3      3
4      0     5     2         10     10      5      0

========== AVERAGES ==========
Average Waiting Time    : 4.50
Average Turnaround Time : 8.50
Average Response Time   : 1.25

========== CPU UTILIZATION ==========
CPU Busy Time  : 16
CPU Idle Time  : 0
Total Time     : 16
CPU Utilization: 100.00%
```

6.7 Compare All Algorithm

```
Choose Algorithm:
1. FCFS
2. SJF (Non-Preemptive)
3. Priority (Non-Preemptive)
4. SRTF (Preemptive SJF)
5. Priority (Preemptive)
6. Round Robin
7. Compare All Algorithms
0. Exit
Enter choice (0-7): 7

================ ALGORITHM COMPARISON ================
Algorithm                      Avg WT      Avg TAT       Avg RT    CPU Usage
FCFS                             3.50         7.50         3.50      100.00%
SJF Non-Preemptive               2.50         6.50         2.50      100.00%
Priority Non-Preemptive          4.25         8.25         4.25      100.00%
SRTF                             2.50         6.50         2.50      100.00%
Priority Preemptive              6.00        10.00         3.75      100.00%
Round Robin                      4.50         8.50         1.25      100.00%

Round Robin Time Quantum: 2
```

7. Performance Comparison

The CPU Scheduling Simulator compares six scheduling algorithms based on Average Waiting Time (WT), Average Turnaround Time (TAT), Average Response Time (RT), and CPU Utilization. Round Robin uses a time quantum of 2.

Test Case 1

Algorithm	Average WT	Average TAT	Average RT	CPU Utilization
FCFS	5.33	10.00	5.33	100.00%
SJF Non-Preemptive	4.33	9.00	4.33	100.00%
Priority Non-Preemptive	5.33	10.00	5.33	100.00%
SRTF	3.00	7.67	0.00	100.00%
Priority Preemptive	5.00	9.67	3.33	100.00%
Round Robin	5.33	10.00	1.00	100.00%

Test Case 2

Algorithm	Average WT	Average TAT	Average RT	CPU Utilization
FCFS	5.00	9.00	5.00	100.00%
SJF Non-Preemptive	2.75	6.75	2.75	100.00%
Priority Non-Preemptive	6.00	10.00	6.00	100.00%
SRTF	2.75	6.75	2.75	100.00%
Priority Preemptive	6.00	10.00	6.00	100.00%
Round Robin	5.25	9.25	2.50	100.00%

Test Case 3

Algorithm	Average WT	Average TAT	Average RT	CPU Utilization
FCFS	3.50	7.50	3.50	100.00%
SJF Non-Preemptive	2.50	6.50	2.50	100.00%
Priority Non-Preemptive	6.00	10.00	6.00	100.00%
SRTF	2.50	6.50	2.50	100.00%
Priority Preemptive	6.00	10.00	3.75	100.00%
Round Robin	4.50	8.50	1.25	100.00%

8 Performance Analysis

Based on the actual results obtained from the CPU Scheduling Simulator:

1. Lowest Average Waiting Time: SJF Non-Preemptive and SRTF achieved the lowest average waiting time of 2.50 units in Test Case 3.

2. Lowest Average Turnaround Time: SJF Non-Preemptive and SRTF achieved the lowest average turnaround time of 6.50 units in Test Case 3.

3. Lowest Average Response Time: SRTF achieved the lowest average response time of 0.00 units in Test Case 1.

4. CPU Utilization: The CPU utilization was 100% in all three test cases, indicating that the CPU remained busy throughout execution.

5. Effect of Preemption: Preemption allows a running process to be interrupted when another process needs the CPU according to the scheduling algorithm. SRTF reduced waiting and turnaround times in the tested cases.

6. Effect of Round Robin Time Quantum: A time quantum of 2 units allowed each process to receive CPU time in turns. It provided a response time of 1.00 unit in Test Case 1, 2.50 units in Test Case 2, and 1.25 units in Test Case 3.

9. Conclusion

The CPU Scheduling Simulator was successfully developed in C to implement six CPU scheduling algorithms. It accepts process details, displays Gantt charts, and calculates Waiting Time, Turnaround Time, Response Time, their averages, and CPU utilization.

Three test cases were used to compare the performance of the algorithms. The results showed that SJF and SRTF achieved lower waiting and turnaround times in the tested cases, while Round Robin provided CPU time to processes in turns. Overall, this project helped in understanding how different CPU scheduling algorithms work and how they affect process execution and system performance.